



DIABETES RISK

THE DROPOUT'S MANUAL

Everything behind this diabetes screening tool — the disease, the machine, and every page of the app — explained from zero, with no jargon left standing.

CONTENTS

Five parts. Read them in order the first time — each one assumes the one before it.

CONTENTS P.3

How to read this 3

PART ONE / UNDERSTANDING DIABETES P.4

What diabetes actually is 5

Why it goes unnoticed 6

The eight measurements 7

Why there are two front doors 8

What the percentage means 10

The honest limits 11

PART TWO / THE PARTS OF THE MACHINE P.12

Four pieces, one job each 13

The knowledge base 15

The engine 16

The interface 17

How they talk to each other 18

PART THREE / WHAT HAPPENS WHEN YOU SUBMIT P.19

The one-sentence version 20

Step 1 — validate 22

Steps 2 and 3 — score and judge 23

Steps 4 to 7 — write and send 24

Three doors — and what is not happening 26

PART FOUR / THE CLINICAL REFERENCE PAGE P.28

What the page is and why 29

Measurements one to three 30

Measurements four to six 31

Measurements seven and eight 32

Probabilities into instructions 33

Guidance, context, disclaimer 34

PART FIVE / THE MODEL CARD P.35

A nutrition label for the model 36

The five big numbers 37

The threshold dial 38

Where it gets things wrong 40

The curve of every threshold 41

What the forest leans on 42

How it was trained 43

Where it should not be trusted 44

The part for programmers 45

REFERENCE P.46

Appendix A — Jargon buster 46

Appendix B — The whole thing on one page 47

HOW TO READ THIS

This manual exists because the app's own pages are written for people who already speak the language. This one assumes you speak none of it.

THE FIVE PARTS

01 Understanding diabetes

What the disease actually is, and what the eight measurements mean. Nothing about computers.

02 The parts of the machine

Every folder and file in the project, what it does, and how they talk to each other.

03 What happens when you submit

One person's numbers, followed from the button press to the answer. This explains the **About** page.

04 The clinical reference page

Every medical number the tool uses, and who says so. This explains the **Clinical reference** page.

05 The model card

How good the model really is, and where it fails. This explains the **Model card** page.

THE ONE SENTENCE TO REMEMBER

If you take nothing else from these pages, take this: **the forest does the arithmetic, the knowledge base supplies the facts, and the AI only does the writing.** Almost every design decision in the project follows from that single rule.

A NOTE ON THE NUMBERS

Every figure in this manual was read from the running system, not estimated. The timings were measured on a real submission; the accuracy scores come straight out of the trained model's own report.

01

UNDERSTANDING DIABETES

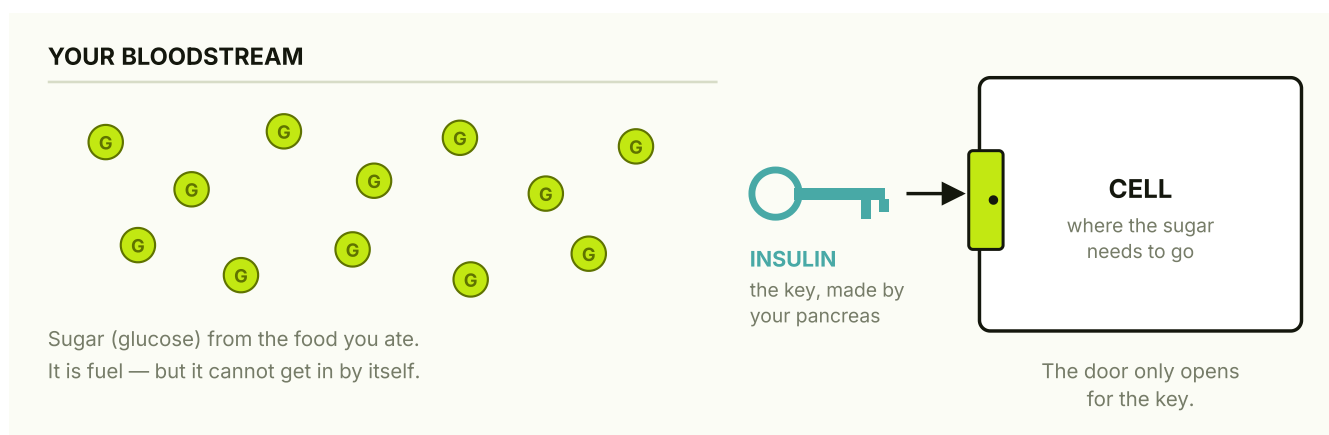
Before any of the code makes sense, the disease has to. Start here even if you think you already know what diabetes is — the mechanism is the part everyone gets wrong.

FOUNDATIONS

WHAT DIABETES ACTUALLY IS

When you eat, your body breaks the food down into **sugar** — glucose — and puts it into your blood. That sugar is fuel. But it cannot get into your cells on its own. It needs a key to unlock the door.

That key is a hormone called **insulin**, made by an organ called the pancreas. Every time sugar arrives in your blood, your pancreas releases insulin, the doors open, the sugar goes in, and your blood returns to normal.



Sugar needs insulin to unlock the cell. Diabetes is this system breaking down.

DIABETES IS THAT SYSTEM BREAKING

There are two ways it can break, and they are genuinely different diseases that happen to share a name.

TYPE 1

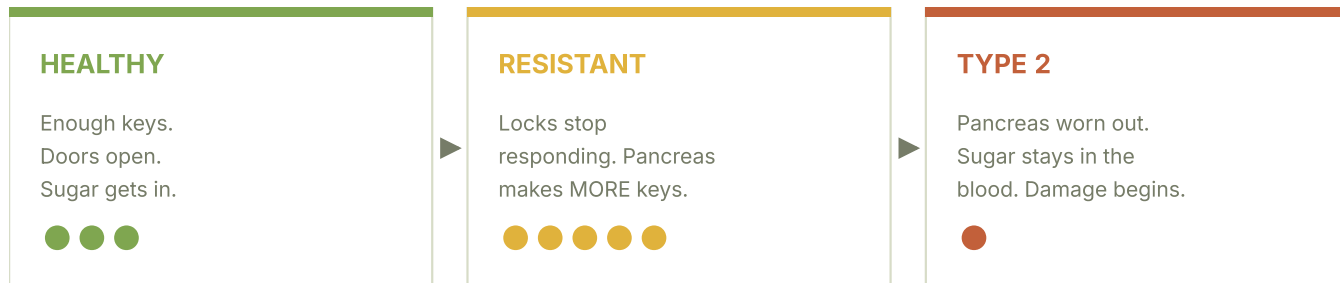
The pancreas stops making the key altogether. It usually begins in childhood, comes on fast, and is not caused by lifestyle. **This app is not about Type 1.**

TYPE 2 — WHAT THIS APP SCREENS FOR

The body still makes keys, but **the locks stop responding to them**. So the pancreas makes more keys, and more, trying to force the doors open. For years it manages to keep up. Then it wears out.

Either way, the result is the same: the sugar cannot get in, so it stays in the blood. And too much sugar sitting in your blood for too long quietly damages things — your eyes, your kidneys, your nerves, your feet, your blood vessels.

WHY IT GOES UNNOTICED FOR YEARS



Type 2 develops slowly, over years, without symptoms.

Notice what is happening in the middle panel. The pancreas is working **harder than normal**, and it is winning. Blood sugar still looks fine on a test. The person feels completely healthy. Nothing hurts.

That stage can last **ten years or more**.

THE CRUEL PART

Type 2 diabetes does not hurt. There is no pain, no obvious symptom, nothing that sends you to a clinic. Most people only find out once something has **already been damaged** — blurred vision, a foot wound that will not heal, a kidney problem found by accident.

THAT SILENCE IS THE ENTIRE REASON SCREENING TOOLS EXIST.

A screening tool goes looking for the disease in people who feel perfectly fine. It catches the thing while it is still quiet — and, crucially, while it is still largely reversible. Losing five to seven percent of your body weight at the middle stage can genuinely stop the slide.

WHY THIS MATTERS IN NIGERIA

Africa has the **highest proportion of undiagnosed diabetes of any region in the world**. Most people who have it never find out. Labs are scarce outside the big cities. A tool that runs on measurements a small health centre can already take is worth building for exactly that reason.

THE EIGHT THINGS THE MODEL LOOKS AT

The model learned from **768 real patients**. For each one it was told these eight measurements, plus whether that person actually turned out to have diabetes. From that, it learned which patterns tend to go together.

#	MEASUREMENT	IN PLAIN TERMS
1	Glucose	How much sugar is sitting in your blood. By far the most important one.
2	BMI	Whether your weight is high for your height. Extra body fat is what makes the locks stop responding.
3	Age	Risk climbs as you get older. The pancreas tires out, regardless of weight.
4	Family history	If your parents or siblings have it, you are considerably more likely to.
5	Insulin	How many keys your body is pumping out. Lots of keys plus high sugar means the locks are failing.
6	Pregnancies	Pregnancy strains the insulin system. More pregnancies, more strain.
7	Blood pressure	Often breaks down alongside the same underlying insulin problem.
8	Skinfold thickness	A pinch of fat on the back of the arm — another way to gauge body fat.

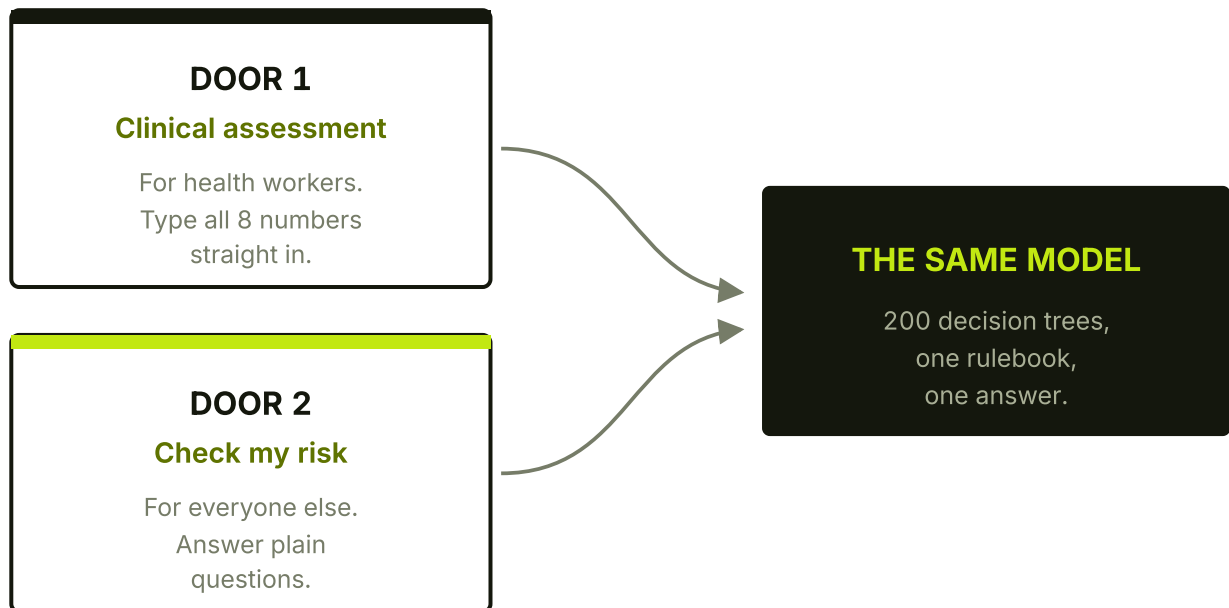
A USEFUL WAY TO PICTURE IT

Think of the model as a **very experienced nurse** who has personally seen those 768 patients, remembers how each one turned out, and can now recognise the shape of a risky profile at a glance. She cannot diagnose you. But she can tell you, with good instincts, whether you ought to go get tested.

Part Four of this manual goes through all eight in full clinical detail — what counts as normal, what counts as worrying, and which health authority set each number.

WHY THE APP HAS TWO FRONT DOORS

Here is the problem. Look back at that list of eight. **Glucose, insulin and skinfold thickness can only come from a clinic.** You cannot know your insulin level sitting at home. So a single page cannot serve everybody.



Two front desks. One brain behind them.

DOOR 2 HAS TO DO EXTRA WORK

The interview asks things a normal person can actually answer — how old are you, how tall, how heavy, has anyone in your family had diabetes — then converts those answers into the eight numbers the model needs:

- Height and weight become **BMI**, which is just a formula.
- Family history becomes a **family-risk score**, using real values taken from the original patient data.
- Insulin and skinfold are **not asked at all**.

THAT LAST DECISION WAS TESTED, NOT ASSUMED

Hiding insulin and skinfold from the model changed its accuracy by **0.002**. Essentially nothing — it barely leans on them. So asking would mean pestering someone for a number they cannot possibly know, in exchange for no benefit.

Blood sugar is different. That one genuinely matters, so the interview asks "have you ever had it measured?" If yes, excellent. If no, the check still works — just less sharply. It catches roughly 69 at-risk people in 100 instead of 78.

Both doors feed the exact same model. Same brain, different front desk.

WHAT THE PERCENTAGE ACTUALLY MEANS

This is the part almost everyone gets wrong, so it is worth reading twice.

IT DOES NOT MEAN WHAT YOU THINK

If the app says **83%**, that does **not** mean “you have an 83% chance of getting diabetes.”

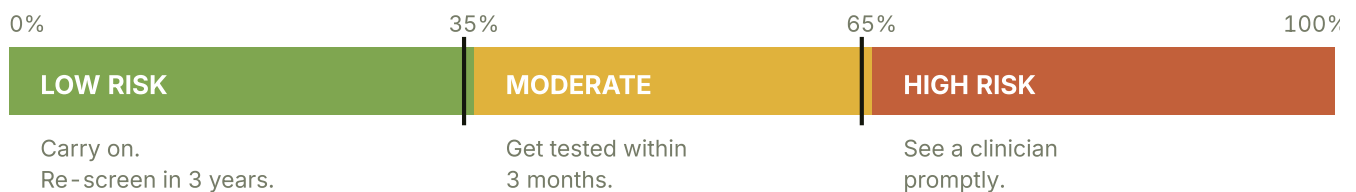
What it actually means is this:

OUT OF THE ORIGINAL PATIENTS WHOSE NUMBERS LOOKED LIKE YOURS, ABOUT 83% TURNED OUT TO HAVE DIABETES.

It is a **resemblance score**. It measures how much your profile looks like the ones that turned out badly. That is a reason to go and get a blood test. It is not a diagnosis, and it cannot be one — only a laboratory test can diagnose diabetes.

THE NUMBER BECOMES AN INSTRUCTION

A bare percentage is useless to a person, so the app sorts it into one of three bands, and each band carries a specific action:



The model's number becomes an instruction.

Notice that each band names **the exact test to ask for**. That is deliberate. “See a doctor” is advice nobody acts on. “Ask for an HbA1c” is something you can walk into a clinic and say out loud.

THEN IT EXPLAINS ITSELF

The app also writes you a short plain-English explanation of which of your numbers pushed the score up, and what to do next. How that explanation gets written — and why an AI is allowed nowhere near the number itself — is Part Three.

THE HONEST LIMITS

A tool that will not tell you where it fails cannot be trusted where it works. These are the four things that genuinely limit this one.

01 THE TRAINING PATIENTS WERE NOT NIGERIAN

The 768 patients were **adult women of Pima Indian heritage, in the United States**. Diabetes risk works somewhat differently in Nigerian populations — different genetics, diet and environment. Treat the number as a nudge to get tested, not as a personal forecast.

02 768 PEOPLE IS NOT MANY

It is a small dataset by any standard. Every accuracy figure in this manual has real wobble in it. Read “78%” as “roughly 78%”.

03 ALL THE TRAINING PATIENTS WERE WOMEN

Which means **“number of pregnancies” carries no information for a man**. The tool is on weaker ground for male users, and says so on its own model card rather than hiding it.

04 IT FINDS A RESEMBLANCE, NOT A DISEASE

The model learned which patterns *went together with* a positive test. That is a statistical association. It cannot detect diabetes, cannot rule it out, and does not replace a laboratory glucose or HbA1c measurement.

WHY THE LIMITS ARE PRINTED AT ALL

Most student projects show only the flattering number. This one has an entire page devoted to its own failures — including the twelve at-risk people it missed out of 154. That honesty is not modesty. In anything medical, a tool that oversells itself is **more dangerous than no tool at all**.

02

THE PARTS OF THE MACHINE

The project is not one program. It is four separate pieces that each do one job and hand their work to the next. Once you see the pieces, the whole thing stops looking complicated.

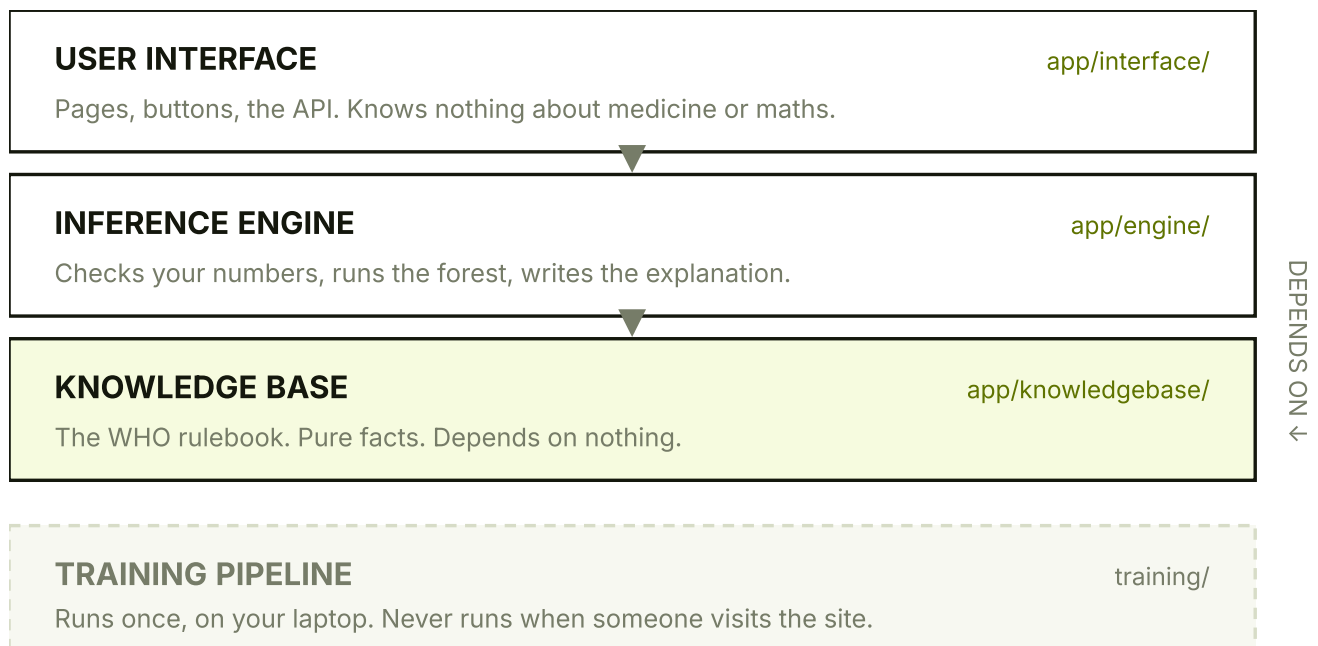
ARCHITECTURE

EXPLAINS: ABOUT › THREE LAYERS

FOUR PIECES, ONE JOB EACH

Imagine a small clinic. There is a **receptionist** who greets you and hands you a form. A **nurse** who takes your numbers and works out what they mean. And a **medical textbook** on the shelf that both of them consult, but neither of them wrote.

That is exactly how this project is arranged.



Each layer may only lean on the one below it. Nothing ever reaches upward.

THE RULE THAT HOLDS IT TOGETHER

Each layer may only lean on the layer **below** it. Nothing ever reaches upward. The textbook does not know the receptionist exists.

That sounds like a technicality. It is the single most useful property of the whole codebase, and here is why:

- You can **correct a medical threshold** without touching any page, button, or model.
- You can **redesign the entire website** without any risk of changing a clinical number.
- You can **test the nurse without building a clinic** — the engine runs perfectly well with no website attached, which is how the automatic tests check it.

THE TRAINING PIPELINE IS THE ODD ONE OUT

Notice the dashed box. **The training code never runs when someone visits the site.** It runs once, on your laptop, and its only output is a file — `model.pkl`, the trained forest, frozen and saved to disk. The live app just reads that file. Think of it as the difference between *teaching* the nurse and *asking* the nurse.

THE KNOWLEDGE BASE: THE TEXTBOOK

The bottom layer. Pure facts, no thinking. This folder contains every medical number the app will ever use, and it imports nothing from anywhere else.

FILE	WHAT IT HOLDS
<code>who_idf_thresholds.json</code>	The rulebook itself. Every cut-off, every band, every piece of official guidance, the risk bands, and the legal disclaimer. Just data — no instructions.
<code>clinical_reference.py</code>	The librarian. Reads that file and answers questions like <i>“a glucose of 168 — which band is that?”</i>
<code>prompt_builder.py</code>	Assembles the fact sheet that gets handed to the AI, so the AI never has to remember a medical number.

WHY THIS IS THE MOST IMPORTANT DESIGN CHOICE IN THE PROJECT

An AI will happily tell you “diabetes starts at 200 mg/dL.” It sounds confident. But it is recalling that from memory, and memory can be wrong — it might say 180, or 220, and it would sound **exactly as confident either way.**

So the AI is never asked to remember. All the numbers live in one file, and three different parts of the app read from that same file: the page that displays the rules, the code that judges your readings, and the AI, which is handed the facts before it is allowed to write a word.

Change the number once, and all three change together. They cannot drift apart, because there is only one copy.

There is a date stamped in that file — `last_reviewed` — so a future reader can tell how stale the rulebook might be. Medical guidelines get updated; this records when someone last checked.

THE ENGINE: THE NURSE

The middle layer, and where all the actual thinking happens. Five files, each with one clear job. Crucially, **none of them knows a website exists**.

FILE	ITS ONE JOB
<code>validation.py</code>	The bouncer. Checks every number is real, sane, and inside the allowed range before anything else is allowed to touch it.
<code>predictor.py</code>	The scorer. Loads the trained forest, runs your numbers through it, and reads each value against the rulebook.
<code>interview.py</code>	The translator. Turns plain answers — height, weight, family history — into the eight numbers the model expects.
<code>narrative.py</code>	The writer. Talks to the AI. Also contains a backup writer that needs no internet at all.
<code>chat.py</code>	The receptionist who speaks freely. Pulls numbers out of a normal sentence and calls the scorer.

WHY IT MUST NOT KNOW ABOUT THE WEBSITE

Because that is what lets the engine be tested on its own, thousands of times, in under five seconds, with no browser involved. It is also why the same engine could be dropped into a phone app or an SMS service without rewriting a single clinical rule.

THE TWO WRITERS

Look again at `narrative.py`. It contains **two** ways to write your explanation:

- A **plain writer** that builds sentences directly from the rulebook. A little robotic, never fails, needs no internet.
- An **AI writer** that reads better, needs a working connection, and is allowed to fail.

The plain one always runs. The AI one is attempted afterwards. Part Three shows exactly how they hand over.

THE INTERFACE: THE RECEPTIONIST

The top layer. Pages, buttons, forms, colours. It orchestrates everything and understands none of it — and that is on purpose.

PIECE	WHAT IT IS
<code>routes.py</code>	The switchboard. When a browser asks for <code>/model-card</code> , this decides what to fetch and which page to build.
<code>api.py</code>	The same thing for machines instead of people — returns raw data for other programs to use.
<code>templates/</code>	The page layouts. Skeletons with blanks that get filled in with real values.
<code>static/css/</code>	Every colour, size and spacing rule. Hand-written, no framework.
<code>static/js/</code>	The small amount of code that runs inside your browser: the chatbot, the step-by-step interview, the dark-mode switch.

THE RULE YOU CAN CHECK YOURSELF

There is not a single medical number anywhere in this layer. No threshold, no cut-off, no band boundary. If you searched every template for the number 140, you would not find it — the page asks the knowledge base and prints whatever comes back.

TEMPLATES, IN ONE PICTURE

A template is a page with holes in it. The route fetches the real values, hands them over, and the template fills its own blanks:

```

Template says:  "Your risk is {{ percentage }}%"
Route provides: percentage = 82.5
Browser receives: "Your risk is 82.5%"

```

The template never calculates. It only displays.

HOW THEY ACTUALLY TALK TO EACH OTHER

Four layers is the theory. Here is the conversation that happens when a real person clicks a real button.

1 Browser → Interface

"Somebody submitted this form. Here are eight numbers."

2 Interface → Engine

"Are these numbers valid?" The interface does not know what valid means. It just asks.

3 Engine → Knowledge base

"What range is glucose allowed to be in?" The engine does not know either. Only the rulebook knows.

4 Engine → the saved model file

"Score this person." Back comes one number: 0.825.

5 Engine → Knowledge base, again

"0.825 — which band? And a glucose of 168, what is that?"

6 Engine → Interface

A finished result object: the percentage, the band, all eight readings judged, and the explanation.

7 Interface → Browser

A finished page. Done.

READ THE ARROWS AGAIN

Every single arrow points **downward or back up the same path**. The knowledge base never asks the engine for anything. The engine never asks the interface for anything. Information flows down as questions and back up as answers — never sideways, never in a loop.

That is what stops a project this size from turning into a knot where changing one thing quietly breaks three others.

03

WHAT HAPPENS WHEN YOU **SUBMIT**

One real person's numbers, followed from the button press to the answer on screen. Every timing here was measured on the running system.

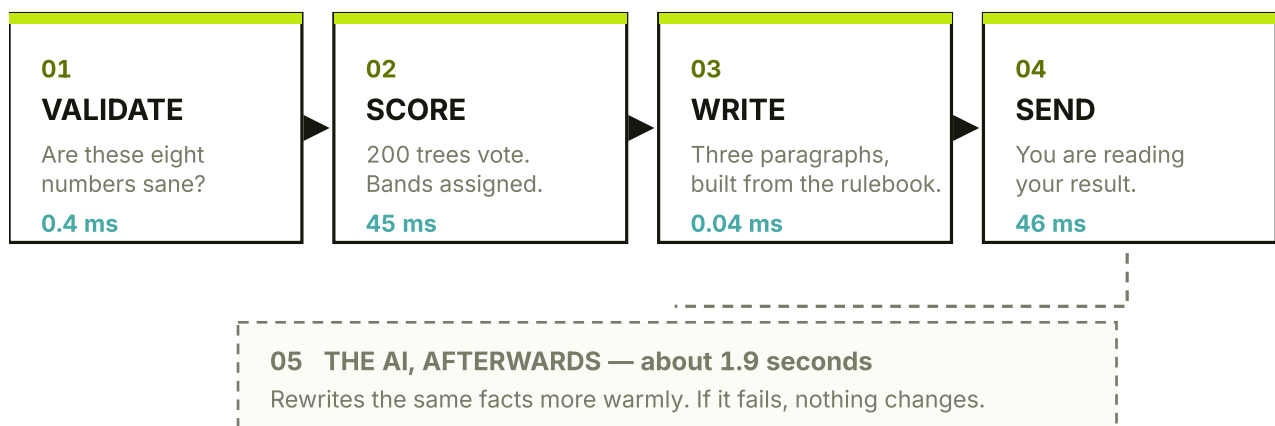
THE JOURNEY

EXPLAINS: ABOUT › REQUEST FLOW

THE ONE-SENTENCE VERSION

Your numbers get checked, scored by the forest, and the page comes back to you in **about a twentieth of a second** — before the AI has been involved at all. The AI's version arrives roughly two seconds later and quietly replaces the text.

That two-stage delivery is the thing the About page never quite explains, and it is the key to the whole design.



Everything that matters is done in 46 ms. The AI is a garnish, added last.

LET US FOLLOW ONE REAL PERSON

These eight values were pushed through the actual system to produce every number in this chapter.

Pregnancies	3	Insulin	180
Glucose	168	BMI	34.5
Blood pressure	88	Family history	0.62
Skin thickness	32	Age	47

THE ANSWER

82.5% — High Risk. The next four pages are every step it took to get there.

STEP 0 — THE NUMBERS LEAVE YOUR COMPUTER

You press the button. Your browser bundles the eight values and posts them to the server. Nothing has been calculated yet — your browser has no idea what the answer is, because **it does not contain the model**. All the thinking happens on the server.

STEP 1 – VALIDATE

0.4 MILLISECONDS

The bouncer at the door. Before anything touches the model, every value is checked. It is stricter than you would expect, and each rule earns its place.

THE CHECK	WHY
Is it actually there?	A blank field gets <i>"This measurement is required."</i>
Is it a number?	You typed <code>abc</code> or <code>12kg</code> — rejected. Digits and one decimal point only.
Is it believable?	The allowed range is read from the same rulebook file as the medical thresholds — so the form and the guidance can never disagree.
Whole where it must be?	Nobody has 3.5 pregnancies or is 47.2 years old. Those two demand whole numbers.
Is it a sane length?	Nothing over 24 characters. If someone pastes a 10,000-character string to jam the server, it is thrown out before any work happens.
Commas become dots	Someone typing <code>34,5</code> meant 34.5. Common in many countries, so it is accepted rather than punished.

The code deliberately rejects things a computer would otherwise accept quite happily — `infinity`, `nan`, hexadecimal, scientific notation like `1e5`. A nurse would never type those, so if they appear, something is wrong.

IF ANYTHING FAILS, YOU DO NOT LOSE YOUR WORK

The form comes back with the message pinned to **the exact field that was wrong**, and everything you already typed still sitting in place. Nobody should have to re-enter eight numbers because they fat-fingered one.

Our person passes all six checks. No errors.

STEP 2 – SCORE 45 MILLISECONDS

Your eight numbers are laid out in a single row — like one line of a spreadsheet — and handed to the forest.

All 200 trees vote. Each one walks your numbers down its own set of questions (*is glucose above 154? is BMI above 29?*) and lands on a yes or a no. Then the votes are counted.

About 165 of the 200 said yes for this person. That is your **0.825 — 82.5%**.

A SMALL DETAIL THAT PREVENTS A SILENT DISASTER

The row is labelled with column *names*, not just thrown in as eight numbers in order. The pipeline internally shuffles the columns around. If the code relied on position, **your glucose could be read as your BMI** — and nobody would notice, because a number would still come out. It would just be the wrong one.

STEP 3 – JUDGE EACH MEASUREMENT

The forest gives one number. But 82.5% does not tell you *why*, so the system now reads each of your eight values against the rulebook:

Glucose 168 → falls in 140–199 → **Impaired Glucose Tolerance**, amber

It does this for all eight, then sorts them **most serious first**. For our person, seven of the eight came back flagged.

Then it does one more thing. It crosses your flagged readings against **how much the model actually relies on each one**, producing your top drivers:

1. GLUCOSE 2. BMI 3. AGE

That is the difference between “*seven things are off*” and “*these three are what is driving your score.*” Your blood pressure being high matters less to the result than your glucose does — and the page tells you so.

STEPS 4 TO 7 – WRITE, SEND, IMPROVE

STEP 4 – WRITE THE EXPLANATION 0.04 ms

Still no AI. The server writes three paragraphs itself, slotting your actual result into sentence shapes built from the rulebook. Plain, faintly robotic, and **it can never fail**. No internet needed, no API key.

STEP 5 – THE PAGE IS SENT 46 ms total

Faster than you can blink. You now have a complete result page: your percentage, your band, all eight readings judged, your top drivers, and a written explanation. **If the internet died at this instant, you would still have everything you need.**



STEP 6 – THE AI JOINS IN, QUIETLY about 1.9 s

Now that you are already reading your result, a second request goes out in the background asking for a better-written version. Of what gets sent, note carefully what the AI is and is not given:

- Your probability and band — **already calculated. It cannot change them.**
- Your top drivers, and which band each fell into.
- **The actual guideline text**, lifted straight out of the rulebook.
- Instructions: plain English, three paragraphs, never introduce a number you were not given, never diagnose.

**THE AI IS NOT WORKING OUT YOUR RISK.
IT IS A WRITER HANDED A FACT SHEET.**

STEP 7 – THE SWAP

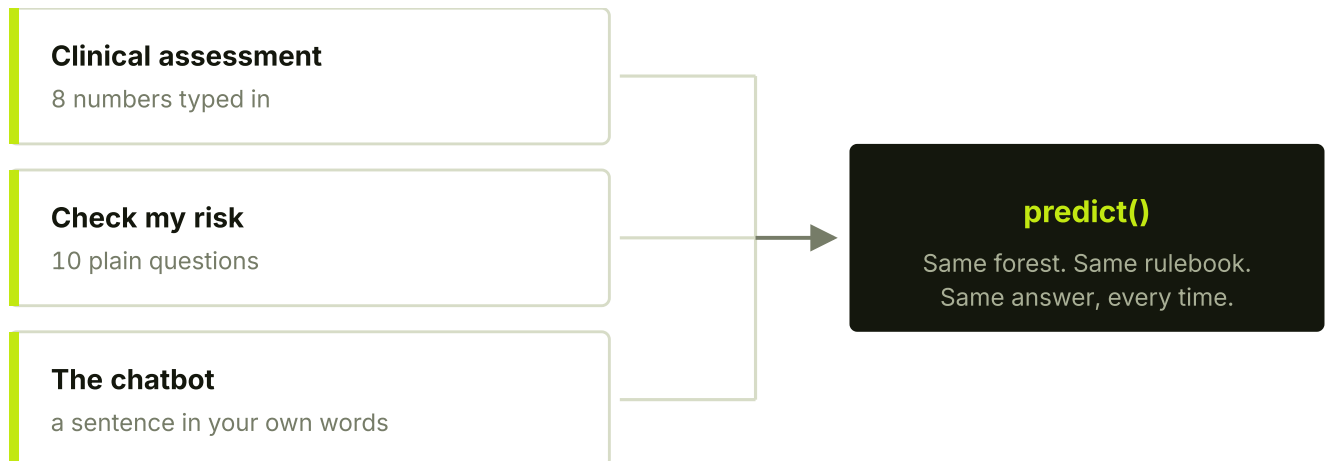
The new text fades in and replaces the robotic version, and a small line notes that the AI wrote this one.

AND IF IT FAILS?

No key configured, provider down, rate limit hit, network drops, request too slow — then **nothing happens at all**. The original text simply stays. You would never know anything had been attempted. That is the entire reason for doing it in this order.

THREE DOORS, ONE ROOM — AND WHAT IS *NOT* HAPPENING

Part One described two front doors. There are actually three, and this is the machinery underneath all of them.



Three ways in. All of them end at the same function.

The interview adds one step before scoring: your plain answers become the eight numbers. Height and weight become BMI; "several relatives, diagnosed young" becomes **0.88** on the family score; insulin and skinfold are marked **never measured**.

That last part matters. Those two get filled with a typical value so the model can run — but they are then **left out of your results entirely**. Telling you “your insulin is high” when nobody measured your insulin would be inventing a finding.

The chatbot's only job is to **pull values out of your sentence**. When it has enough, it is not allowed to guess your risk — it must call the same prediction function everything else calls.

WHAT IS DELIBERATELY NOT HAPPENING

Nothing is saved

No database. Your eight numbers exist in the server's memory for about 46 milliseconds and are then gone.

No logging of your values

Errors get written to the log. Your health data does not.

Nothing is cached

Responses are marked `no-store`, so your result cannot get stuck in a shared cache on a clinic computer and shown to the next person.

Refresh and it is gone

Forever. That is not a bug. There is genuinely nowhere to look it up.

THE SENTENCE THAT SUMS UP THE WHOLE PROJECT

The forest does the arithmetic. The knowledge base supplies the facts. The AI only does the writing.

Every design choice follows from that. The AI never sees a number it could get wrong, because the number was calculated before it was called. And it is called *last*, and *optionally*, so it can fail without taking anything down with it.

04

THE CLINICAL REFERENCE PAGE

Every medical number this tool uses, where it came from, and what it means for a person. This is the app's rulebook, printed for humans.

THE RULEBOOK

EXPLAINS: CLINICAL REFERENCE

WHAT THE PAGE IS, AND WHY IT EXISTS

It is **the rulebook**. Every single number this app uses to decide “that reading is normal” or “that reading is worrying” is written down on that page, along with who says so.

WHY PRINT IT AT ALL?

Because a health tool that will not show its sources is asking to be trusted on faith. Publishing the thresholds means a nurse, a lecturer, or an examiner can check them against the actual WHO documents — and can catch you if a number is wrong.

ONE FILE, THREE READERS

All these numbers live in a single file. The page you are reading, the code that judges your values, and the AI that writes your explanation **all read the same file**. They cannot drift apart, because there is only one copy.

That is what the page means when it says *“Nothing is recalled from a language model's memory.”*

THE PARAGRAPH AT THE TOP

“Cut-off values used to interpret the 8 clinical parameters... These are screening -interpretation aids, not diagnostic criteria.”

Translation: **these numbers help you decide who to send for a test. They do not tell you who has diabetes.** Only a lab can do that. The sentence is there so no health worker mistakes the page for a diagnostic manual.

HOW TO READ EVERY ENTRY

Each of the eight measurements gets the same four things: what it is, its coloured bands, the official rule, and the source. The bands are the useful part — a traffic light:

NORMAL	PREDIABETES	DIABETES RANGE
under 140 mg/dL	140 – 199 mg/dL	200 and above mg/dL

Every parameter works like this: green is fine, amber is watch it, red is a problem.

The row of buttons near the top is just navigation. The page is long; clicking “Glucose” jumps you to that section.

THE MEASUREMENTS, ONE TO THREE

01 PREGNANCIES

Under 5 typical · 5 or more high parity, amber

Pregnancy can cause a temporary diabetes that goes away after birth — but women who have had it are far more likely to develop permanent type 2 diabetes later. More pregnancies means more chances of that having happened. WHO says those women should be screened **every three years for life**.

02 GLUCOSE the important one

Under 140 normal · 140–199 prediabetes, amber · 200+ diabetes range, red

This is the closest thing to looking at the disease directly, because diabetes is literally defined as too much sugar in the blood.

The test works like this: you drink a measured sugary solution — 75 grams of glucose — wait two hours, then have blood taken. A healthy body has cleared that sugar away by then. A diabetic body cannot, so it is still floating around.

ONE READING IS NEVER ENOUGH

The page states that a single high result **must be confirmed on a different day** before anyone is diagnosed. People have bad days; labs make mistakes.

03 BLOOD PRESSURE

Under 80 normal · 80–89 high-normal, amber · 90+ hypertensive, red

When your BP is read as “120 over 80”, this is the 80 — the pressure in your arteries while your heart rests between beats.

Why is it in a diabetes tool? Because both often come from the same underlying problem: **insulin resistance**. So they travel together. High blood pressure does not cause diabetes, but it is a strong hint the same machinery is misbehaving.

The page is honest about a limitation: the dataset recorded only the bottom number, so a full hypertension classification is not possible from it.

THE MEASUREMENTS, FOUR TO SIX

04 SKINFOLD THICKNESS

Under 30mm typical · 30–45mm elevated, amber · 45mm+ high, red

A nurse pinches the skin at the back of your upper arm with a caliper — a small clamp — and measures the fold. It estimates body fat.

Why bother when BMI already exists? Because **where** you carry fat matters, not only how much. Two people can weigh the same and have quite different risk depending on where it sits.

Worth saying in your write-up: the page states outright this is **not a WHO criterion**. It is a supporting clue, and the source line says so.

05 INSULIN

Under 30 low, amber · 30–166 typical · above 166 hyperinsulinaemia, red

Notice something odd — **both too low and too high are flagged**. That is not a mistake:

- **Too low** means your pancreas is not making enough keys.
- **Too high** means your pancreas is screaming, pumping out keys, but the cells have stopped listening. That is insulin resistance — the early stage of type 2.

THE SNEAKY ONE

In the high-insulin case, **your glucose can still look completely normal**. The pancreas is compensating. It is an early warning that shows up *before* blood sugar goes wrong — which is exactly the window a screening tool wants to catch.

06 BMI

<18.5 under · 18.5–25 healthy · 25–30 over · 30/35/40+ obesity I, II, III

Weight in kilos divided by height in metres squared. Six bands — the most of any measurement.

The single most useful sentence on the whole page lives here: **losing 5–7% of your body weight measurably reduces the chance of prediabetes becoming diabetes**. For a 100kg person that is 5–7kg. Not a transformation — a realistic change that actually works.

There is also a deliberate decision recorded: WHO has *lower* BMI cut-offs for Asian populations. The page states plainly that the standard cut-offs apply to African populations and the Asian ones must not be substituted.

THE MEASUREMENTS, SEVEN AND EIGHT

07 DIABETES PEDIGREE FUNCTION

Under 0.4 limited · 0.4–0.9 moderate, amber · above 0.9 strong, red

The strangest one, and worth understanding properly because an examiner will ask about it. **It is not a medical test.** It is a number someone invented in 1988 to squash “how much diabetes runs in your family” into a single figure. It weighs two things:

- **How closely related** — your mother counts far more than a cousin.
- **How young they were diagnosed** — a parent diagnosed at 30 is a much worse sign than one diagnosed at 75, because early onset suggests stronger genetics.

The page is upfront: *“It is specific to this dataset, not a laboratory test.”*

THIS IS THE NUMBER THE INTERVIEW HAS TO ESTIMATE

A normal person cannot compute it. So plain answers are mapped onto real values taken from the original patient data:

No family history	0.17
One parent or sibling	0.37
Several close relatives	0.63
Several, diagnosed young	0.88

08 AGE

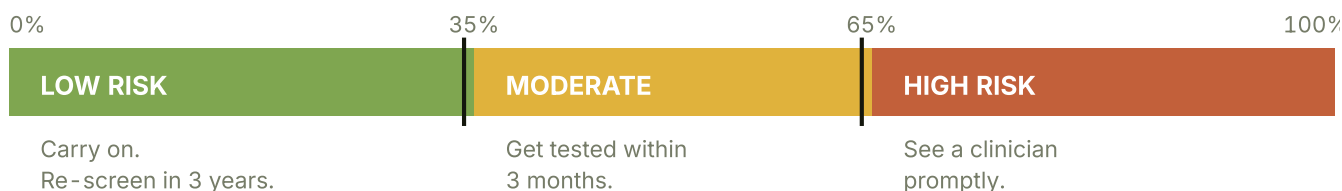
Under 35 green · 35–44 amber · 45 and above red

Nothing you can do about it, but it matters: the cells that make insulin gradually wear out with age, regardless of weight.

Which is why routine screening is recommended **from 45** for everyone — and **earlier** if you are overweight, hypertensive, or have family history.

TURNING A PROBABILITY INTO AN INSTRUCTION

The model returns a number between 0 and 1 — say 0.62. That means nothing to a person. Three bands turn it into something you can act on.



The model's number becomes an instruction.

BAND	RANGE	WHAT YOU DO
Low	0–35%	Keep doing what you are doing. Re-screen every 3 years, or yearly from 45.
Moderate	35–65%	Get a fasting glucose or HbA1c test within 3 months. Start lifestyle changes now.
High	65–100%	See a clinician promptly. Ask specifically for a fasting plasma glucose, an OGTT, or an HbA1c.

WHY EACH BAND NAMES THE ACTUAL TEST

“See a doctor” is advice nobody acts on — it is vague, and it puts the burden on a worried person to work out what to ask for. **“Ask for an HbA1c” is a sentence you can say out loud at a clinic desk.** That difference decides whether a screening result turns into an actual test.

A WORD ON THE BOUNDARIES

Notice they are not evenly spaced, and they do not match the 34.3% threshold the model uses internally to decide who to flag. Those are two different jobs:

- The **threshold** is the tool's own yes/no line, tuned to catch as many real cases as possible. Part Five covers it in full.
- The **bands** are how a human should read the number, and are about urgency of action rather than a decision.

GUIDANCE, CONTEXT, AND THE DISCLAIMER

FOUR THINGS TRUE FOR EVERYONE

01 Confirm before concluding

No screening score is a diagnosis, ever. WHO requires a lab-confirmed result, repeated on a separate day, before diabetes can be diagnosed.

02 Lifestyle change comes first

150 minutes of moderate exercise a week, sugar under 10% of your calories, and losing 5–7% of body weight. This roughly **halves** the rate at which prediabetes becomes diabetes — a bigger effect than most drugs.

03 Tobacco and alcohol

Smoking raises the risk of type 2 diabetes and speeds up the damage to blood vessels once you have it.

04 Undiagnosed diabetes causes silent damage

Years without symptoms while the eyes, kidneys and nerves take damage. **This is the reason screening exists at all.**

THE NIGERIA SECTION

Why this project makes sense in this specific place: millions of Nigerian adults have diabetes, and **Africa has the highest share of undiagnosed cases of any region** — most people never find out. Labs are scarce outside big cities, so a tool running on measurements a basic health centre already has is genuinely useful. Urbanisation and dietary change are pushing rates up across West Africa.

WORTH POINTING OUT TO AN EXAMINER

The source line says the figures are **indicative and should be checked against the current Atlas before citing**. That is the difference between a serious project and one that invents statistics.

THE DISCLAIMER

An educational screening aid. It does not diagnose, does not replace a clinician, and **must not be used to start, stop or change any treatment**. Anyone with symptoms or an elevated result should see a licensed professional.

05

THE **MODEL** CARD

How good the model actually is, measured on patients it had never seen — and an honest account of everywhere it falls short.

THE REPORT CARD

EXPLAINS: MODEL CARD

A NUTRITION LABEL FOR THE MODEL

Food packaging has to tell you the calories and the sugar, whether or not the answer is flattering. A **model card** does the same for a machine learning model: what it was fed, how well it works, and — crucially — where it fails.

The idea comes from AI ethics research and is now standard practice for any honest project. Most student work shows only the good number. Yours has an entire page for the bad ones.

THE LINE UNDER THE TITLE

Diabetes Risk Random Forest v1.0.0 · RandomForestClassifier · trained 2026-08-26

A **random forest** is the type of model. Here is the idea, and it is genuinely simple:

TWO HUNDRED OPINIONS, ONE VOTE

Imagine asking **one** doctor whether you are at risk. They might have a quirk — perhaps they over-weight family history.

Now ask **200 doctors**, each trained on a slightly different mix of patients, and each allowed to look at only a few of your measurements. Then take a vote.

The individual quirks cancel out. The shared signal survives. That is a random forest: 200 simple decision-makers — “trees” — voting. If 130 of the 200 say “at risk”, your probability is 0.65.

WHY EVERY NUMBER IS MEASURED ON STRANGERS

All the figures on that page come from **154 records the model had never seen**. You train on 614, hide 154 away, then test on those.

It is the difference between a student who memorised the past papers and one who can answer new questions. Testing on data it trained on would only measure memory.

THE FIVE BIG NUMBERS

0.809

ROC-AUC

The headline score. 0.5 is guessing, 1.0 is perfect.

78%

SENSITIVITY

Of people who really have diabetes, how many we catch.

70%

SPECIFICITY

Of healthy people, how many we correctly clear.

73%

ACCURACY

Everyone we got right. This number lies — see below.

WHAT ROC-AUC ACTUALLY MEANS

More concrete than people expect:

PICK ONE PERSON WITH DIABETES AND ONE WITHOUT, AT RANDOM. HOW OFTEN DOES THE MODEL SCORE THE DIABETIC ONE HIGHER?

80.9% of the time. That is your 0.809. The beauty of it is that it does not depend on where you set your cut-off — it measures the model's ability to *rank* people, which is a property of the model itself.

WHY ACCURACY IS LISTED FOURTH, NOT FIRST

500 of the 768 patients do not have diabetes. So a “model” consisting of the single word “no” would score **65% accuracy**. It would be completely useless and catch precisely zero cases — but the number looks respectable. That is why sensitivity and specificity are shown separately: accuracy alone hides the failure.

THE FIFTH NUMBER: IS IT STABLE?

Cross-validated AUC 0.839 ± 0.020

Instead of one split, chop the training data into 5 pieces and run 5 times, each time holding out a different piece:

0.861 0.825 0.863 0.812 0.832

That tightness is the point. If one fold scored 0.86 and another 0.61, the result would depend on a lucky split. It does not. The model is **genuinely this good, not fortunate**.

THE DIAL THAT WAS DELIBERATELY MOVED

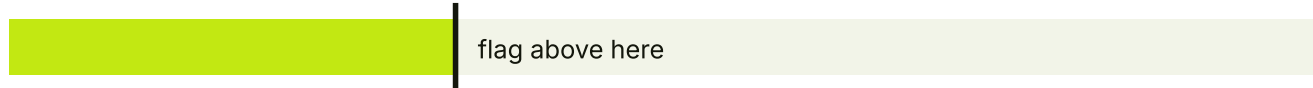
The default rule would be: above 50% probability, flag them. Below, clear them. **That is the wrong rule for screening** — and the proof is in the model's own numbers.

THE TEXTBOOK DEFAULT — 50%



Catches only 54% of real cases. Misses nearly half of them.

WHAT THIS TOOL USES — 34.3%



Catches 78% of real cases. Costs more false alarms — worth it.

Lowering the bar catches more sick people. That is the whole trade.

WHICH MISTAKE COSTS MORE?

Miss a diabetic person

They walk away reassured, get no treatment, and the silent damage carries on for years. Possibly irreversible.

These are not remotely equal. So you **lower the bar** — flag anyone above 34.3% instead of 50%. The threshold was lowered until at least 80% of true cases were being caught.

Flag a healthy person

They go for a blood test, it comes back clean, and they have lost an afternoon.

METRIC	AT 0.500 DEFAULT	AT 0.343 IN USE
Sensitivity	0.5370	0.7778 ↑
Specificity	0.8700	0.7000 ↓
Precision	0.6905	0.5833 ↓
F1 score	0.6042	0.6667 ↑
Accuracy	0.7532	0.7273 ↓

Read it as a **trade, not an upgrade**. Accuracy even goes down — and that was accepted on purpose, because catching sick people matters more than looking good.

THE SUBTLE BIT THAT KEEPS IT HONEST

The cut-off was chosen using **training data only**. The 154 test records were never consulted while picking it. Tune the dial by peeking at the test set and your scores are flattered and meaningless — you have secretly used the answer key.

WHERE IT GETS THINGS WRONG

Four boxes, covering every possible outcome for the 154 test patients. This is the most honest picture of the model that exists.

	PREDICTED LOW	PREDICTED AT RISK
ACTUALLY HEALTHY	70 Correctly cleared	30 Sent for a test they did not need
ACTUALLY DIABETIC	12 MISSED — the costly error	42 Correctly flagged

All 154 test patients. The 12 in red are the ones that matter.

READING IT

- **70** healthy people, told they are fine. Correct.
- **42** diabetic people, flagged. Correct — and this is the win.
- **30** healthy people sent for a test they did not need. Wasteful, and completely recoverable.
- **12** diabetic people **told they were fine**.

THE TWELVE

That last box is the one that matters. Twelve people walked away from a tool that should have caught them. The page says it plainly: **the second error is recoverable; the first is not**. A wasted blood test can be undone. A missed diagnosis costs years.

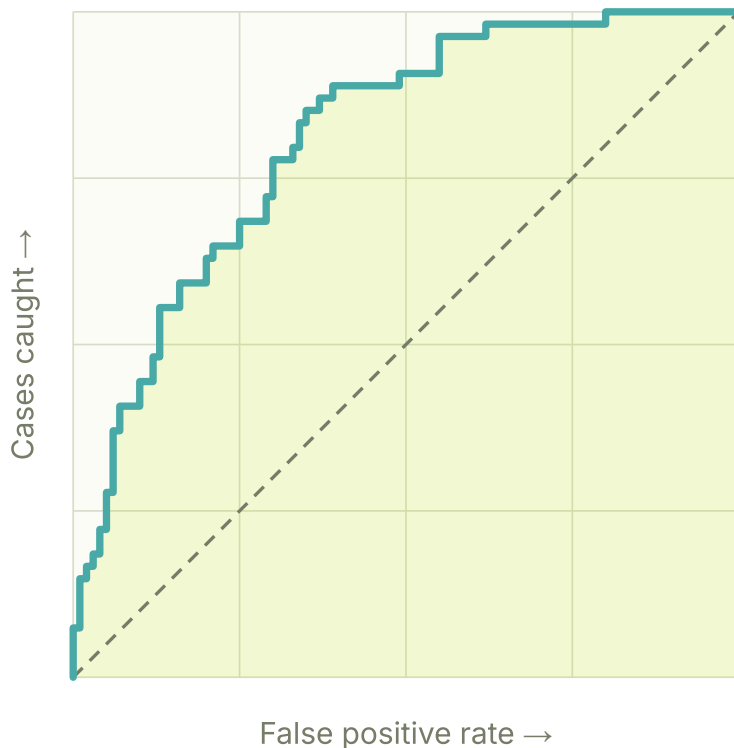
TWO WORDS YOU WILL SEE NEXT TO THESE

Precision — when the tool shouts “at risk”, how often is it right? 58%. So roughly 4 in 10 alarms are false. That is acceptable for a tool whose job is to say *go get tested*, not *you are ill*.

F1 — a single score blending precision and sensitivity, for when you want one number instead of two. It improved when the threshold moved, which is a decent sanity check that the trade was sensible overall.

THE CURVE OF EVERY POSSIBLE THRESHOLD

Remember the dial from the last chapter? This one graph shows **every possible setting of it at once**.



AUC 0.809

Your model.

0.500

A coin flip
(the dashed line).

The further the curve
bulges up and left,
the better.

Every possible setting of the threshold dial, drawn at once.

Up the side is how many real cases you catch. Along the bottom is how many false alarms you cause.

Each point on the curve is one possible threshold. Drag the dial one way and you catch more but cause more false alarms; drag it back and the reverse. The curve traces that entire trade-off in a single line.

HOW TO READ IT IN THREE SECONDS

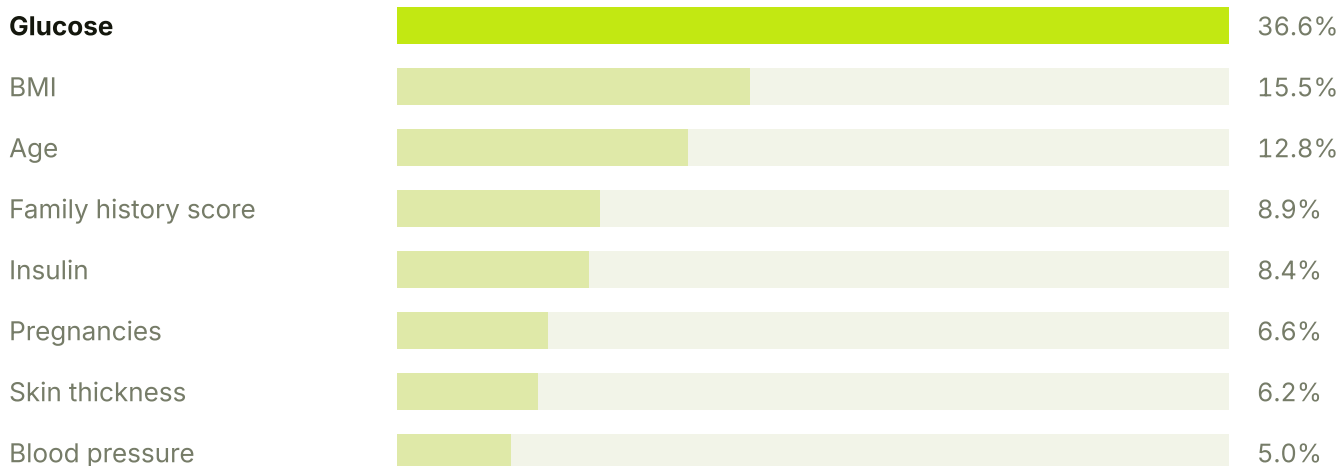
The dashed diagonal is a **coin flip**. A useless model sits exactly on it. The further your curve bulges **up and to the left**, the better it is — because that corner means catching lots of cases while raising few false alarms.

The shaded area underneath is literally the AUC. Your 0.809 means the curve covers 80.9% of the box. A perfect model would fill it completely.

This is the same 0.809 from the headline numbers — the graph is just the picture of it, rather than another measurement.

WHAT THE FOREST ACTUALLY LEANS ON

Eight measurements go in. They are not treated equally — and the model decided the weighting entirely on its own.



Nobody programmed this. The model worked out on its own that glucose matters most.

THIS IS THE BEST EVIDENCE THAT THE MODEL IS SANE

Nobody programmed that ordering. The model worked out from the data alone that glucose matters most — and landed on exactly the measurement WHO uses to *define* the disease. Weight, age and family history following behind is what any doctor would tell you.

That is a real check, not decoration. If blood pressure had come out on top, something would be badly wrong — either a bug, or the model latching onto a coincidence. **The model independently rediscovered the textbook.**

Insulin ranking fifth is lower than clinical intuition suggests, and there is an honest reason for it: nearly half the insulin readings in the original data were never taken. A model cannot lean on a measurement that is half invented. Chapter 29 covers that in full.

This same ranking is what the result page uses to tell you which of *your* readings drove *your* score — it crosses this list against whichever of your values came back flagged.

HOW IT WAS TRAINED

The data

Pima Indians Diabetes Dataset · 768 records · 8 measurements · 500 without diabetes, 268 with · split 614 train / 154 test.

Stratified split

Both halves got the same disease ratio. Split carelessly and you could land 90% of the diabetics in one half, making every number meaningless.

The settings

200 trees · no tree deeper than 6 questions · never decide on fewer than 4 people · each tree sees only a random handful of the measurements. Chosen by trying **72 combinations** and keeping the best.

Random seed 42

Retrain tomorrow and you get these exact numbers again. Anyone can verify your results.

THE TRAP HIDDEN IN THIS DATASET

In the original data, **a missing measurement was recorded as 0**. But a BMI of zero is impossible, and a glucose of zero means you are dead. Those zeros mean *“nobody measured this”*, not *“the value is zero.”*

Feed them in raw and the model learns nonsense. So they are replaced with the middle value of that column — and fitted on **training data only**, so nothing about the test set leaks backwards. Same discipline as the threshold.

WAS IT JUST MEMORISING?

The model scores about **0.14 higher** on data it studied than on data it has not. Some gap is normal — everyone does better on material they have seen. A gap this size on 768 rows is acceptable. A gap of 0.4 would mean it had memorised rather than learned.

The page also lists exact library versions. Boring, and necessary: tools change behaviour between releases, and without that list someone reproducing your work in two years cannot tell whether you were wrong or their software moved.

WHERE THIS MODEL SHOULD NOT BE TRUSTED

Five admissions. This is the most valuable section in the whole project, and the part a good examiner will respect most.

01 THE TRAINING POPULATION IS NOT THE TARGET POPULATION

Adult Pima Indian women in the United States. The tool is for Nigeria. Different genetics, diet, environment. Real deployment would need locally collected data. **This is the biggest limitation, and it is listed first rather than buried.**

02 IT IS A SMALL DATASET

768 records, and a test set of only 154. A handful of records changing would visibly move every number. Treat “78% sensitivity” as roughly -78, not exactly -78.

03 NEARLY HALF THE INSULIN READINGS WERE MISSING

374 of 768 insulin values and **227** skinfold values were never measured. Filling them with the median makes the data usable but flattens the real variation — which is *why insulin ranks fifth* when a doctor would put it higher. The model cannot value a measurement that is half invented.

04 IT PREDICTS AN ASSOCIATION, NOT A DIAGNOSIS

It learned which patterns went together with a positive test in one group of people. That is correlation. It cannot detect diabetes and cannot rule it out.

05 SEX AND PREGNANCY ARE ENTANGLED

Trained only on women. So for a man, “Pregnancies” is a constant zero fed into a slot the model expects to vary. Weaker ground — and it says so.

WHY PUBLISHING THIS MAKES THE TOOL BETTER, NOT WORSE

A screening tool is a **prompt to get tested**. Everything above argues for exactly that reading and against treating the percentage as a verdict. Stating the limits is what makes the honest use of the number possible.

THE PART FOR OTHER PROGRAMMERS

The last section of the model card is a small table of web addresses. It is there so this tool can be plugged into something else — a hospital system, a phone app, an SMS service — without anyone reusing the code by hand.

ADDRESS	TYPE	WHAT IT GIVES BACK
<code>/health</code>	GET	Is the AI up? Actually pings the provider rather than just checking a setting.
<code>/api/health</code>	GET	Full readiness: the model, the AI, and the knowledge base.
<code>/api/predict</code>	POST	Send eight numbers, get the probability and band. No AI involved — same answer every time.
<code>/api/narrative</code>	POST	The same, plus the written explanation.

TWO WORDS WORTH KNOWING

GET means “give me something” — what your browser does when you visit a page. **POST** means “here is some data, do something with it” — what happens when you submit a form.

WHY A HEALTH CHECK EXISTS AT ALL

The AI runs on someone else's service. That service can go down, run out of free allowance, or quietly retire the model you were using. **Without a health check you would only find out through a user complaining that explanations stopped appearing.**

`/health` makes a real call to the provider and reports what actually happened — not merely whether a key is present in a settings file. There is a real difference between *configured* and *working*, and only one of them matters.

GOOD ERRORS ARE A FEATURE

Send bad data and you get back a message naming **which field** was wrong, not just “error”. The same discipline as the form telling a nurse exactly which box to fix.

JARGON BUSTER

Every technical word in this manual, in one place.

Accuracy

Everyone you got right, out of everyone. Misleading on its own.

AUC / ROC-AUC

How well the model ranks a sick person above a healthy one. 0.5 is guessing.

Band

A range of values with a label: normal, watch, problem.

Cross-validation

Running the test five times on different splits to check the score was not luck.

Confusion matrix

The four-box table of right and wrong answers.

Endpoint

A web address a program can talk to.

F1

One number blending precision and sensitivity.

Feature

One measurement the model uses. This model has eight.

Hyperparameter

A setting you choose before training — how many trees, how deep.

Imputation

Filling in a missing value with a sensible stand-in.

Leakage

When the answer accidentally hides in the question. Makes scores look brilliant and mean nothing.

Model

The trained pattern-recogniser. Here, a saved file called `model.pkl`.

Overfitting

Memorising the training data instead of learning from it.

Precision

When it shouts “at risk”, how often it is right.

Probability

A number from 0 to 1 for how much you resemble the diabetic patients.

Random forest

200 simple decision-makers voting together.

Sensitivity / recall

Of the truly sick, how many you catch. **The number that matters most here.**

Specificity

Of the truly healthy, how many you correctly clear.

Threshold

The cut-off line for flagging someone. Here, 34.3%.

Training / test split

Hiding some patients away so you can mark the model on unseen work.

THE WHOLE THING ON ONE PAGE

If you have to explain this project in two minutes, say these things in this order.

1 THE PROBLEM

Type 2 diabetes has no symptoms for years while it does damage. Africa has the highest proportion of undiagnosed cases in the world, and labs are scarce outside cities.

2 THE QUESTION

Can eight measurements a small health centre can already take be turned into a reliable prompt to go and get tested?

3 THE ANSWER

Yes, to a useful degree. **ROC-AUC 0.809** on patients it had never seen, catching **78%** of real cases.

4 THE DESIGN RULE

The forest does the arithmetic. The knowledge base supplies the facts. The AI only does the writing — it is never allowed to produce or recall a number.

5 THE HONESTY

The threshold was moved from 50% to 34.3% on purpose, trading accuracy for caught cases. The model card publishes the 12 people it missed as prominently as the 42 it caught. And the training patients were not Nigerian, which is stated first among the limitations rather than buried.

NUMBER WORTH MEMORISING	VALUE
Patients trained on	768
Held-out test patients	154
ROC-AUC on unseen data	0.809
Cases caught (sensitivity)	78%
Screening threshold	34.3%
Trees in the forest	200
Time to a full result	46 ms